

Design of a Pole-Changing Wound Field Synchronous Machine With Bipolar Optical Power Transfer Excitation

Enes Ayaz¹, *Graduate Member, IEEE*, Nail Tosun², *Graduate Member, IEEE*

¹Department of Electric Power and Energy Systems, KTH Royal Institute of Technology, Stockholm, Sweden

² Department of Electrical and Electronic Engineering, Newcastle University, Newcastle, United Kingdom

This paper presents the electromagnetic design and analysis of a pole-changing wound field synchronous machine (PC-WFSM) supplied by a bipolar optical power transfer (OPT) field-excitation system. The proposed rotor excitation architecture uses three stationary light sources: two optical channels provide reversible excitation for rotor field-coil groups whose current polarity changes between pole modes, while a third channel supplies the field-coil groups that retain a fixed polarity. In each bipolar channel, two rotor-mounted photovoltaic (PV) cell groups are selectively illuminated; the inactive PV group provides a passive return path through its first cell and an external diode. A unified PV-cell model is used to describe both the illuminated power-generating regime and the dark forward-biased return regime. Based on this model, a corrected maximum power point (MPP) matching condition is derived and incorporated into the rotor field-winding design, thereby coupling the OPT parameters to the achievable rotor magnetomotive force (MMF), field resistance, slot geometry, and stack length. Two-dimensional finite element analysis of a 10 kW, 4-pole/2-pole prototype is used to verify the pole transformation, rotor-MMF reconfiguration, airgap flux distribution, no-load EMF, torque production, and loss performance. The proposed method provides brushless bipolar field excitation for PC-WFSMs with stationary-side optical control and without rotor-side active switching or resonant compensation components.

Index Terms—Bipolar field excitation, brushless excitation, electrically excited synchronous machine (EESM), finite element analysis (FEA), optical power transfer (OPT), photovoltaic cell, pole-changing, wound field synchronous machine (WFSM).

I. INTRODUCTION

WOUND field synchronous machines (WFSMs) have attracted renewed interest as rare-earth-free alternatives to permanent-magnet machines in traction and power-generation applications [1]. Their controllable field excitation enables wide-speed-range operation, flux weakening, and safe de-excitation under fault conditions. These advantages are especially attractive when high-speed operation and controllable airgap flux are required.

Pole-changing operation provides an additional degree of freedom by modifying the effective pole number to reshape the torque–speed characteristic over a wide operating range [2]. In induction machines, the rotor field naturally follows the stator pole number; in synchronous machines, however, the rotor excitation pattern must also be reconfigured. Hence, a PC-WFSM requires not only a pole-changing stator MMF but also a rotor field system capable of changing the spatial polarity of selected field coils. This requirement makes bipolar field excitation a central design constraint rather than an auxiliary power-supply feature, as illustrated in Fig. 1.

Several brushless excitation methods have been proposed to avoid slip rings and brushes, but they offer different compromises in terms of rotor complexity, controllability, and polarity reversal. Sub-harmonic self-excitation [3] can realize brushless operation by inducing current in a rotor harmonic winding and rectifying it through a diode bridge. However, this approach requires additional rotor windings and rectifiers and does not directly provide reversible field current for pole-changing operation. WPT-based excitation [4] eliminates mechanical contacts, but it generally requires rotor-mounted resonant networks, rectifiers, and filtering compo-

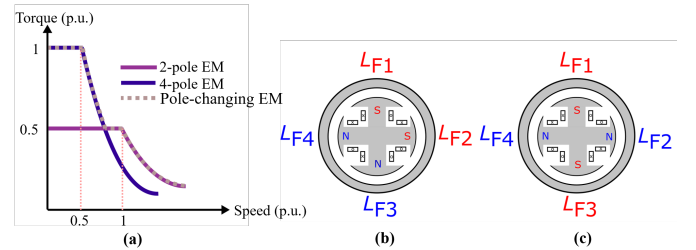


Fig. 1. Pole-changing operation in WFSMs. (a) Torque–speed characteristics for 4-pole and 2-pole modes. (b) and (c) Transition between modes requires reversal of the field current polarity in selected rotor coils.

nents. For bipolar excitation, primary-side controlled WPT has been shown to reverse the field current for pole-changing EESMs [5]; nevertheless, the receiver still includes resonant magnetic components and switching/rectification circuitry.

OPT-based field excitation provides a different route. In the recent OPT-WFSG design framework [6], a shaft-mounted PV cell directly supplies the rotor field winding, reducing rotor-side passive components and making the approach attractive for high-speed machines. The limitation is that the conventional PV-to-field-winding connection is intrinsically unipolar. Therefore, the open design problem addressed in this paper is how to obtain bipolar field excitation for a PC-WFSM while preserving the rotor simplicity of OPT and, at the same time, incorporating the resulting OPT constraint into the machine electromagnetic design.

In this paper, a bipolar OPT excitation system is proposed for pole-changing WFSMs. Two stationary light sources provide bipolar excitation for the field coil groups requiring polarity reversal, while a third independent light source supplies the field coil groups that remain unipolar. In the bipolar channels,

field-current polarity is selected by the active light source.

The main contributions of this paper are summarized as follows:

- A bipolar OPT excitation architecture for PC-WFSMs, enabling field-current reversal by stationary-side optical selection without rotor-side active switches or resonant compensation components.
- A unified illuminated/dark PV-cell model and a corrected MPP matching condition for bipolar OPT excitation.
- Integration of the corrected bipolar OPT constraint into the rotor field-winding, MMF, and geometry design of a 4-pole/2-pole WFSM.
- FEA verification of the pole transformation, rotor-MMF distribution, airgap flux density, no-load EMF, torque performance, and loss characteristics.

The remainder of this paper is organized as follows. Section II presents the proposed bipolar OPT excitation and PV-cell model. Section III details the machine design procedure. Section IV presents FEA validation. Section V compares the proposed method with existing approaches, and Section VI concludes the paper.

II. PROPOSED BIPOLAR OPT EXCITATION AND PV-CELL MODEL

Fig. 2 shows the proposed field-excitation architecture for the PC-WFSM. The system uses three stationary light sources. LS1 and LS2 supply the rotor field-coil groups whose current polarity must be reversed during pole changing, whereas LS3 supplies the field-coil groups that keep the same polarity. The optical links are converted to electrical field power by rotor-mounted PV cells, so the field excitation is controlled from the stationary side without slip rings, rotor-side active converters, or resonant compensation components.

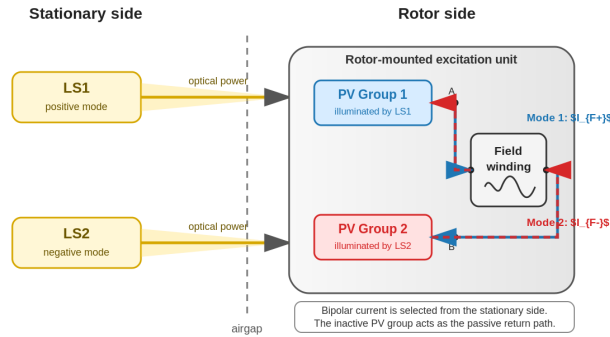


Fig. 2. Conceptual illustration of the proposed bipolar OPT excitation architecture for a PC-WFSM.

The PV cell is represented by the single-diode model

$$I = I_{ph} - I_{D0} \left[\exp \left(\frac{q(V + R_s I)}{nkT} \right) - 1 \right] - \frac{V + R_s I}{R_p} \quad (1)$$

where I_{ph} is the photocurrent, I_{D0} is the diode saturation current, R_s and R_p are the series and shunt resistances, and n is the ideality factor. For a conventional unipolar OPT field winding, the MPP matching condition is [6]

$$R_f = \frac{V_{MPP}}{I_{MPP}}, \quad (2)$$

where V_{MPP} and I_{MPP} are the terminal MPP voltage and current of one illuminated PV cell. For N_s identical series-connected cells, the MPP voltage is $N_s V_{MPP}$, and the current remains I_{MPP} .

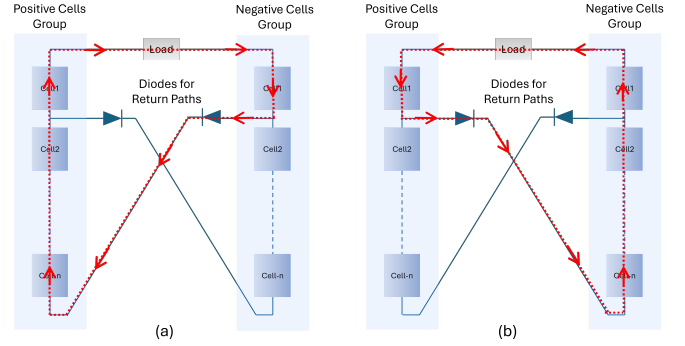


Fig. 3. Operating principle of the proposed bipolar OPT excitation. In each mode, one PV group is illuminated and the first cell of the inactive group forms the passive return path together with an external diode.

The bipolar channel consists of two identical series PV groups, G_+ and G_- , connected to opposite terminals of the field winding, as shown in Fig. 3. When LS1 illuminates G_+ , current flows from Node A to Node B and produces positive field current I_{F+} . When LS2 illuminates G_- , the current direction is reversed, and negative field current I_{F-} is obtained. The current polarity is therefore selected only by the active light source. For symmetric PV groups and return paths,

$$|I_{F+}| \approx |I_{F-}|. \quad (3)$$

A key feature is that the return current passes through an external diode, completing the loop.

The effective field voltage at the MPP is therefore

$$V_{MPP}^* = N_s V_{MPP} - V_{f,return}, \quad (4)$$

where

$$V_{f,return} = V_{f,ext} + I_{MPP} R_{s,return}. \quad (5)$$

The corrected bipolar OPT matching condition becomes

$$R_f^* = \frac{N_s V_{MPP} - V_{f,return}}{I_{MPP}}. \quad (6)$$

This relation is the main difference from unipolar OPT excitation: the field winding must be designed for the PV MPP after subtracting the passive return-path voltage penalty, as shown in Fig. 4.

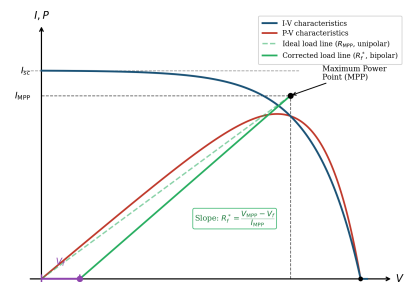


Fig. 4. Unified I-V characteristic illustrating the illuminated generating regime, single-cell dark diode-conduction regime, and load-line shift caused by the passive return path.

The corresponding return-path loss is

$$P_{\text{return}} = V_{f,\text{ext}} I_F + I_F^2 R_{s,\text{return}}. \quad (7)$$

Accordingly, pole changing is obtained by assigning the rotor field coils to bipolar and unipolar groups. The bipolar groups are connected to LS1/LS2 and reverse their current direction between the 4-pole and 2-pole modes, whereas the unipolar groups are supplied by LS3 and retain the same polarity. Table I summarizes the required coil polarities.

TABLE I
ROTOR FIELD-COIL GROUP EXCITATION FOR POLE CHANGING

Field Winding	Channel	4-pole	2-pole
LF1	Unipolar LS3	+	+
LF2	Bipolar LS1/LS2	-	+
LF3	Bipolar LS1/LS2	-	+
LF4	Unipolar LS3	-	-

III. MACHINE DESIGN PROCEDURE

A. Design Constraints and Target Specifications

The proposed pole-changing WFSM is constrained by both electromagnetic requirements and OPT excitation characteristics. The field current is limited by the PV-cell MPP current, and the field winding resistance must satisfy the corrected matching condition in (6). Therefore, the rotor winding, slot geometry, and machine dimensions must be co-designed with the OPT parameters.

Table II summarizes the machine target specifications and PV-cell parameters used in this study.

TABLE II
TARGET SPECIFICATIONS

Parameter	Value	Unit
Rated output power	10	kW
Speed: 4-pole / 2-pole	3000 / 6000	rpm
DC bus voltage	400	V
Stator current density	4.5	A/mm ²
Rotor current density J_r	3.5	A/mm ²
Core material	35PN230	—
Cooling	Natural convection	—
Optical/PV electrical power per channel	30	W
PV cell V_{MPP}	2.4	V
PV cell I_{MPP}	12.5	A
PV cell R_s	0.08	Ω
PV cell n, I_{D0}	1.5, 2×10^{-8}	—, A

The design procedure used in this paper is summarized as follows. First, the target operating points for the 4-pole and 2-pole modes are specified. Second, the rotor-MMF requirement is obtained from the desired airgap flux and output power. Third, the PV-cell MPP parameters and the number of series cells are selected. Fourth, the corrected bipolar MPP condition in (6) is used to determine the allowable field resistance. Finally, the conductor area, number of field turns, rotor slot geometry, and stack length are selected so that the MMF, current-density, and resistance constraints are simultaneously satisfied.

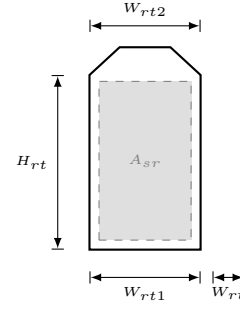


Fig. 5. Rotor slot geometry used for field winding design.

B. Output Power and Induced EMF

The no-load induced electromotive force (EMF) and output power of the machine are expressed as

$$E_0 = k_w \omega_e N_{\text{ph}} \phi_g = \frac{\mu_0 \pi k_w \omega_e N_{\text{ph}}}{P l_g} D_g L_{\text{stk}} N_r I_r \quad (8)$$

$$P_{\text{out}} = \frac{\mu_0 \pi \eta k_w \omega_e \cos \varphi}{P l_g} (D_g L_{\text{stk}}) (N_r I_r) (N_{\text{ph}} I_a) \quad (9)$$

where k_w is the winding factor, ω_e is the electrical angular speed, D_g is the airgap diameter, L_{stk} is the stack length, l_g is the effective airgap length, and $N_r I_r$ is the rotor MMF per pole. Since $N_r I_r$ is constrained by the OPT channel, the achievable machine output is directly linked to the optical excitation capability.

C. Rotor Field Winding Design

The rotor field winding must satisfy both the MMF requirement and the corrected MPP matching condition. The field winding resistance is determined by the rotor slot geometry in Fig. 5 as [6]

$$R_f = \frac{2 P N_r}{\sigma A_{\text{coil}}} \left[L_{\text{stk}} + \left(1 - \frac{\pi}{4} \right) W_{rt} + \frac{\pi}{4} \tau_r \right]. \quad (10)$$

The rotor MMF per pole is

$$N_r I_r = \frac{\sigma I_{\text{MPP}} V_{\text{MPP}}^*}{2 J_r} \left[L_{\text{stk}} + \left(1 - \frac{\pi}{4} \right) W_{rt} + \frac{\pi}{4} \tau_r \right]^{-1}, \quad (11)$$

where the corrected effective voltage is

$$V_{\text{MPP}}^* = N_s V_{\text{MPP}} - V_{f,\text{return}}. \quad (12)$$

The number of turns per slot is

$$N_r = \frac{K_{rf} (W_{rt2} - W_{rt1} - W_{rt}) H_{rt}}{2 A_{\text{coil}}}. \quad (13)$$

Combining (10), (13), and (6), the stack length is obtained as

$$L_{\text{stk}} = \frac{\sigma V_{\text{MPP}}^* A_{\text{coil}}}{J_r K_{rf} (W_{rt2} - W_{rt1} - W_{rt}) H_{rt}} - \left(1 - \frac{\pi}{4} \right) W_{rt} - \frac{\pi}{4} \tau_r. \quad (14)$$

This formulation highlights that the rotor winding resistance and MMF cannot be selected independently but must satisfy the OPT matching constraint.

TABLE III
DESIGNED MACHINE PARAMETERS

Parameter	Value	Unit
<i>Stator</i>		
Outer / inner diameter	200 / 122	mm
Stack length L_{stk}	80	mm
Slots; turns/phase	36; 48	—
Winding factor k_w	0.925	—
<i>Rotor</i>		
Outer diameter; airgap l_g	121; 0.5	mm
Field turns N_f ; slot height H_{rt}	24; 18	—; mm
Field resistance R_f^*	0.093	Ω
Field current I_F ; MMF/pole	12.5; 300	A; At
<i>OPT excitation</i>		
Optical channels	3	—
PV electrical power/channel	30	W
Single-cell return voltage	0.24	V

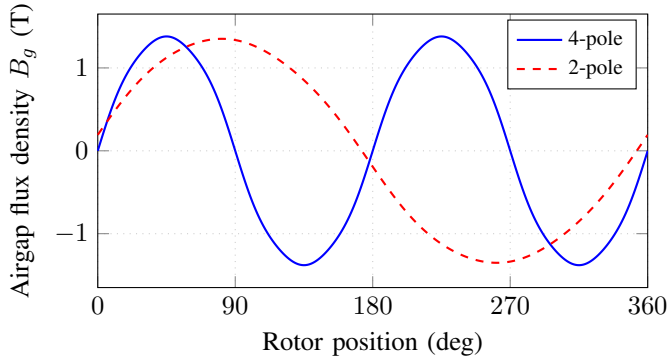


Fig. 6. FEA-computed airgap flux density for the two pole configurations.

D. Final Design Parameters

The design procedure yields the machine parameters summarized in Table III. The final design satisfies the output power requirement, MMF requirement, magnetic saturation limit, and corrected MPP matching condition.

IV. FINITE ELEMENT VALIDATION

A. FEA Setup

Two-dimensional FEA was carried out in ANSYS Maxwell 2024R1 using the nonlinear B-H characteristics of 35PN230 electrical steel. The simulations were performed for both pole configurations, corresponding to the excitation modes described in Section II. No-load and loaded operating points were evaluated to verify the pole transformation, rotor-MMF reconfiguration, flux distribution, EMF waveform, torque production, and loss performance. Core losses were evaluated using the material loss model of the lamination, while copper losses were calculated from the winding resistances and current densities.

B. Airgap Flux Density and No-Load EMF

Fig. 6 shows the airgap flux density distribution for both pole configurations. The 4-pole mode exhibits the expected N-S-N-S pattern, whereas the 2-pole mode produces a single N-S pair, confirming correct pole transformation through field-current reversal.

Fig. 7 presents the corresponding no-load phase EMF waveforms. Both configurations exhibit near-sinusoidal behavior,

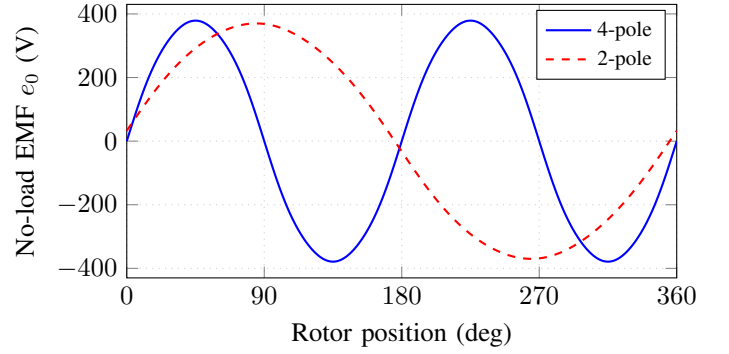


Fig. 7. FEA-computed no-load phase EMF for both pole configurations.

indicating that the pole transition does not significantly degrade the induced voltage waveform.

C. Rotor MMF and Harmonic Content

Because the proposed excitation system is intended to reconfigure the rotor pole number, the rotor-MMF distribution is a key validation result. The field-coil polarities in Table I should be applied to the FEA model and the resulting rotor MMF should be decomposed into spatial harmonics for both pole modes. The desired pole harmonic must dominate in each mode, while the residual unwanted pole harmonic should remain sufficiently small to avoid excessive torque ripple and additional core loss.

D. Torque Production and Torque Ripple

The torque capability of the two pole modes should be evaluated at the same dc-link voltage and current limit used for the drive system. The 4-pole mode is expected to provide higher low-speed torque, whereas the 2-pole mode should extend the high-speed operating range by reducing the electrical frequency. Therefore, the final validation should include average torque, torque ripple, and, if available, torque-speed curves for both modes.

E. Representative Operating Points

To demonstrate the usefulness of pole-changing operation, two representative operating points are evaluated. The first corresponds to a low-speed, high-torque condition, where the 4-pole configuration is expected to be preferable. The second corresponds to a high-speed operating condition, where the 2-pole configuration is expected to reduce electromagnetic frequency and improve efficiency.

TABLE IV
REPRESENTATIVE OPERATING-POINT COMPARISON

Operating point	Speed rpm	Preferred mode	Loss W	Eff. %
Low-speed / high-torque	3000	4-pole	822	92.4
High-speed / reduced-frequency	6000	2-pole	798	93.1

The results in Table IV indicate that the 4-pole mode is suitable for the lower-speed operating point, whereas the 2-pole mode provides improved performance at high speed.

This confirms that the proposed excitation system enables the intended pole-changing functionality from the stationary side.

F. Loss and Efficiency Analysis

Table V summarizes the rated-load loss breakdown for both pole configurations. The dominant losses are stator copper and stator core losses, while the rotor copper loss remains small due to the moderate field current.

TABLE V
LOSS BREAKDOWN AT RATED LOAD

Loss component	4-pole	2-pole	Unit
Stator copper loss	488	488	W
Rotor copper loss	30	30	W
Stator core loss	186	142	W
Rotor core loss	38	28	W
Mechanical	80	110	W
Total machine loss	822	798	W
Machine efficiency	92.4	93.1	%
<i>OPT excitation overhead</i>			
Return-path dissipation	15.5 per active phase		W

The additional loss introduced by the bipolar OPT excitation is associated with conduction through the first cell of the non-illuminated PV group and the external diode, as described by (7). This loss is the main trade-off for eliminating rotor-side active components.

G. Rotor-Side OPT Loss and Thermal Check

The rotor-side excitation loss consists of the field copper loss, the return-path conduction loss, and the PV-cell conversion loss. In the proposed bipolar path, the most critical additional thermal stress is concentrated in the first inactive PV cell that conducts the return current. Therefore, the final design should verify that the dark-cell forward current, external diode current, and local heat generation remain within the allowable temperature limits under continuous excitation.

V. COMPARISON WITH PRIOR METHODS

Table VI compares the proposed excitation system with representative brushless WFSM excitation methods in terms of bipolar capability, rotor-side hardware, pole-changing suitability, and high-speed applicability. Sub-harmonic self-excitation [3] can provide brushless operation, but it requires rotor harmonic windings and diode rectification and does not naturally provide reversible field current. Unipolar OPT [6] offers a simple high-speed rotor structure, but it is limited to fixed-polarity excitation. Primary-side bipolar IPT [5] enables stationary-side current reversal, but still requires WPT coils and receiver-side circuitry. In contrast, the proposed system occupies the previously unfilled intersection of bipolar excitation, primary-side control, and zero rotor-side switching components. Its main trade-off is the return-path voltage drop and associated local loss in the conducting dark PV cell and external diode.

TABLE VI
COMPARISON OF BRUSHLESS EXCITATION METHODS

Feature	[3]	[6]	[5]	Prop.
Transfer method	Sub-harm.	OPT	IPT	OPT
Bipolar field current	No	No	Yes	Yes
Rotor active control	No	No	No ^a	No
Rotor harmonic winding	Yes	No	No	No
Rotor rectifier	Yes	No	Yes	No
Rotor resonant comp.	No	No	Yes	No
Single-cell return path	No	No	No	Yes
Primary-side control	Limited	Yes	Yes	Yes
Pole-changing suitability	Low	Low	High	High
High-speed suitability	Med.	High	Med.	High
Main limitation	Harmonic rotor	Unipolar	Rx circuit	Return loss

VI. CONCLUSION

A pole-changing WFSM with bipolar OPT excitation has been proposed and analyzed. The method enables field-current reversal by using the first cell of the non-illuminated PV group and an external diode as a passive return path, eliminating rotor-side active switches, active converters, and resonant compensation components. The complete excitation architecture uses two optical channels for bipolar field groups and one additional optical channel for unipolar field groups. A unified I–V model was developed for the illuminated and dark PV regimes, leading to a corrected MPP matching condition that links the OPT parameters to the rotor field winding design. FEA results confirmed the intended pole transformation, airgap flux distributions, no-load EMF waveforms, and loss performance in both pole configurations. After the final torque and thermal results are inserted, the proposed topology can be positioned as a brushless bipolar excitation solution for PC-WFSMs, particularly in high-speed applications where rotor-side complexity must be minimized.

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